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Private University established in Karnataka State by Act No. 34 of year 2010
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Project 4

Ethical Considerations in Deep Learning

Exploring bias, privacy, fairness, and transparency in AI systems

Course: Deep Learning

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Date: 06/09/2026

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Introduction

Deep learning has become a huge part of everyday life now, from the recommendations we see on YouTube to the way banks decide who gets a loan, to the filters that decide which resumes even reach a human recruiter. These systems are often praised for being fast, consistent and "data-driven," but along with all these benefits, there are some serious ethical questions that come up when we let algorithms make decisions that affect real people's lives, freedom, money and opportunities.

What makes this topic especially important right now is the scale at which these systems operate. A biased hiring manager might affect a handful of candidates in a week, but a biased hiring algorithm can affect thousands of applications in a single afternoon, all in the exact same unfair way, every single time. That difference in scale is really what turns "a technical limitation" into "an ethical problem" worth taking seriously.

In this project, I looked at three major ethical issues in deep learning, studied a real case where things went wrong, and thought about what could actually be done to reduce these problems going forward. I've also tried to be honest about where these solutions fall short, because I think pretending there's an easy fix would miss the point of the assignment.

Why AI Ethics Matters

Before getting into the specific issues, it's worth asking why this is even a topic that deserves its own field of study, rather than just being treated as a normal engineering bug to fix. The reason is that deep learning systems are increasingly used to make or influence decisions that used to require human judgement — things like who gets bail, who gets interviewed for a job, whose social media post gets flagged, or which patient gets prioritised for treatment.

Unlike a traditional computer program that just follows fixed rules written by a programmer, a deep learning model develops its own internal patterns from data, and often nobody, not even the people who built it, can fully explain every decision it makes. That combination — high-stakes decisions plus low explainability — is exactly why ethics has become such a central part of the conversation around AI, rather than something left only to computer scientists.

A Brief History: How We Got Here

It helps to understand a little bit of the background of how deep learning reached the point where these ethical concerns became unavoidable. For most of the history of computing, software followed rules that a programmer wrote explicitly — if a certain condition was true, the program did a certain thing, and every step could, in theory, be traced back to a line of code someone had written on purpose.

Deep learning flipped that approach. Instead of writing the rules by hand, engineers feed a model huge amounts of example data, and the model gradually adjusts millions of internal numbers (called weights) until it starts producing the right answers on its own. This is what makes deep learning so powerful for tasks like image recognition or language translation, where it would be almost impossible to write explicit rules for every case. But it's also exactly why these systems

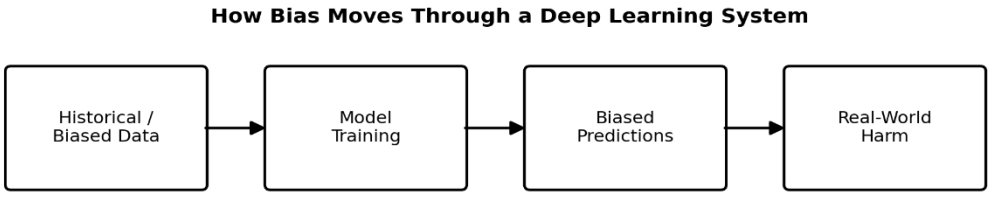
became so hard to audit — nobody sat down and wrote the "rule" the model is using, so nobody can simply read it back out either.

As these models moved from research labs into real products — first into things like spam filters and product recommendations, and later into far more consequential areas like hiring, lending, healthcare and criminal justice — the stakes of getting things wrong grew enormously. A wrong recommendation might mean a bad movie suggestion; a wrong recidivism prediction might mean extra years in prison. This shift in stakes is really what pushed "AI ethics" from a niche academic topic into something that governments, journalists and the general public now pay close attention to.

Task 1: Three Ethical Issues in Deep Learning

1. Bias and Discrimination

One of the biggest problems with deep learning models is that they learn from data, and if that data already has human biases baked into it, the model just ends up copying — and sometimes amplifying — those same biases. This doesn't happen because developers are trying to be unfair; it usually happens because the training data isn't representative of everyone the model will eventually be used on, or because the people labelling the data brought their own unconscious biases into the process without realising it.



There are a few different ways bias can creep into a system. Sometimes it's historical bias, where the data reflects decades of unequal treatment (for example, past hiring records that favour one gender over another). Sometimes it's representation bias, where certain groups simply don't appear enough in the dataset for the model to learn their patterns properly. And sometimes it's what researchers call "proxy" bias, where a feature that seems neutral, like a postal code, ends up correlating strongly with race or income, so the model discriminates indirectly even if it was never given the sensitive attribute directly.

The end result of all of this is that certain groups of people, based on things like race, gender, age or disability, can end up being treated unfairly by a system that is marketed and often trusted as being "neutral" or "objective," simply because it's a machine making the decision instead of a person.

2. Privacy Violation

Deep learning models need massive amounts of data to work well, and a lot of that data is personal — things like photographs, voice recordings, shopping history, location data, or even medical records. Collecting data at this scale already raises questions about consent: did the people whose data was used actually agree to have it used this way, or was it scraped from the internet without their knowledge?

On top of the consent issue, there's a more technical privacy risk. Deep learning models can sometimes "memorise" fragments of the data they were trained on, especially if that data appeared many times or was unusually unique. Researchers have shown that techniques like model inversion attacks or membership inference attacks can be used to reconstruct or confirm pieces of the original training data just by studying how the model responds to certain inputs. This means that even after a model is "finished" and deployed, the private information used to build it hasn't necessarily disappeared.

Facial recognition is probably the clearest real-world example of this issue. Cameras in public spaces, shopping centres, or even school campuses can identify and track individuals without them ever being asked for permission, which raises obvious concerns about surveillance and the erosion of anonymity in everyday life.



3. Lack of Transparency (the "Black Box" Problem)

Most deep neural networks are extremely hard to interpret, even for the engineers who built them. A model might have millions or even billions of internal parameters, and it's often impossible to point to a single, human-readable reason for why it produced a particular output. This is usually called the "black box" problem — data goes in, a decision comes out, and the reasoning in between is essentially invisible.

This becomes a real problem in high-stakes areas like healthcare, hiring, lending, or criminal justice, where someone affected by a decision has every right to know why that decision was made, and to be able to challenge it if it's wrong. If a loan application is rejected or a parole request is denied, "the algorithm said so" is not a satisfying or fair explanation for the person on the receiving end of that decision.

Transparency also matters for accountability. If nobody can explain how a model reached its conclusion, it becomes very difficult to identify where exactly something went wrong when the

system makes a mistake, which in turn makes it harder to fix the underlying issue or hold anyone responsible for it.

A Fourth Issue Worth Mentioning: Accountability

While the task only asked for three issues, I think it's worth briefly mentioning a fourth one that came up repeatedly while I was researching this topic: accountability. When an AI system causes harm — wrongly denies someone a loan, misidentifies someone in a facial recognition search, or contributes to a wrongful arrest — it's often genuinely unclear who is responsible. Is it the company that built the model, the organisation that deployed it, the people who collected the training data, or the individual who chose to trust the output without double-checking it?

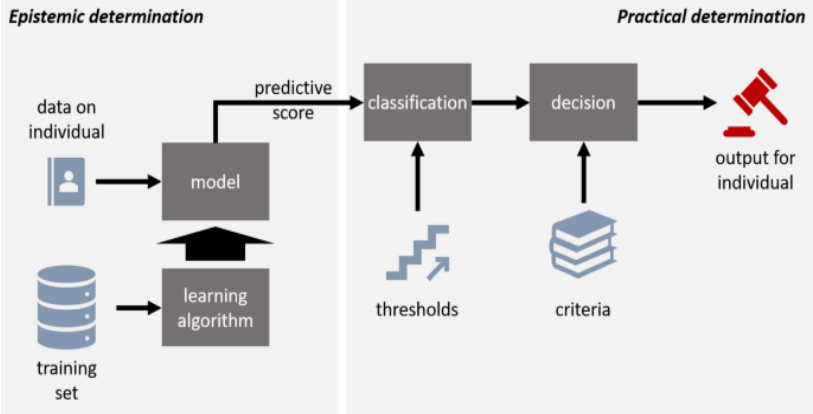
This "accountability gap" makes it harder for affected individuals to get any kind of remedy when something goes wrong, and it can also let organisations avoid responsibility by pointing at "the algorithm" as if it were an independent actor rather than a tool they chose to deploy. I think this issue connects directly back to the transparency problem, since it's very hard to hold someone accountable for a decision that nobody can fully explain in the first place.

Task 2: Real-World Example

A case that really stood out to me while researching this is the COMPAS algorithm, which is used in parts of the US justice system to predict how likely a defendant is to reoffend in the future. Judges in several US states have used COMPAS risk scores to help inform decisions about bail, sentencing and parole, meaning this is not a hypothetical scenario — it's a system that has directly influenced real people's freedom.

In 2016, journalists at ProPublica ran an investigation into COMPAS and found something concerning: the algorithm was much more likely to wrongly flag Black defendants as "high risk" for future crime compared to white defendants, while white defendants who actually did go on to reoffend were more often wrongly labelled as "low risk." In other words, the errors the model made were not evenly spread across racial groups — they consistently disadvantaged one group over another.

The company behind COMPAS pushed back on these findings, arguing that the tool was equally accurate for both groups overall, once you looked at a different statistical measure of fairness. This disagreement points to something deeper than just one company defending its product: there are several different, mathematically valid definitions of "fairness," and it's been shown that a model generally cannot satisfy all of them at the same time.



What makes this example especially useful for this project is that it combines two of the issues from Task 1 at once. It shows bias, because of the unequal error rates between racial groups, and it shows a lack of transparency, because the exact formula COMPAS used to calculate its risk scores was kept private as a trade secret, so defendants couldn't fully understand or challenge the score they were given, even in court. And because this was directly tied to bail and sentencing decisions, it wasn't just a theoretical or academic problem — it affected real people's freedom, sometimes for years of their lives.

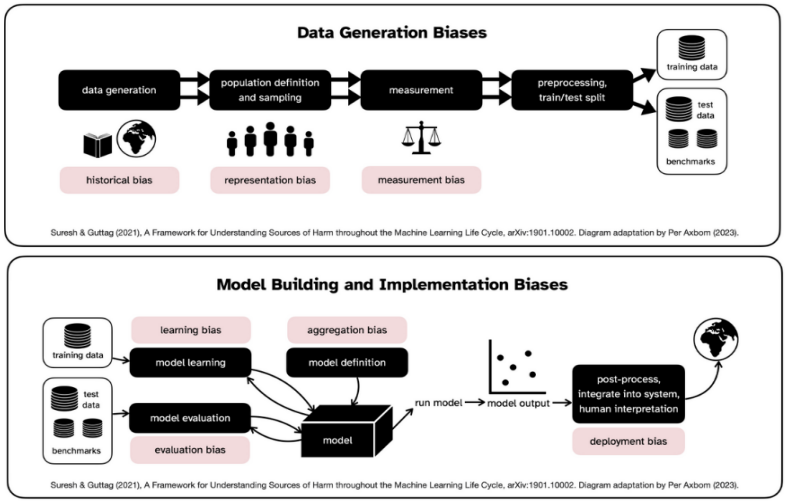
The COMPAS case also shows why "the algorithm is proprietary" is a weak excuse when a system is being used to make decisions about people's rights. If a tool is going to influence something as serious as how long someone stays in prison, there's a strong argument that the public, or at least the courts, should be allowed to inspect how it actually works.

It's also worth noting that COMPAS is far from an isolated incident. Similar concerns have been raised about facial recognition systems that perform noticeably worse on darker-skinned faces and on women, about resume-screening tools that were found to downgrade applications containing the word "women's" (as in "women's chess club"), and about healthcare risk-prediction algorithms that under-prioritised Black patients for extra care because the model used past healthcare spending as a stand-in for medical need, without accounting for unequal access to healthcare in the first place. I chose to focus on COMPAS in detail because it's one of the most well-documented and widely discussed cases, but it's really just one example out of a growing pattern.

Broader Impact on Society and Trust

Cases like COMPAS don't just affect the specific individuals involved in that particular system — they also shape how much the public trusts AI more generally. Once a well-documented case of biased or opaque decision-making becomes public knowledge, it becomes harder for people to trust other AI systems too, even ones that may have been built more carefully. This creates a kind of ripple effect, where the mistakes of one system can slow down the acceptance of genuinely useful AI applications elsewhere, such as in medical diagnosis tools that could otherwise save lives.

On the other hand, I think it's important not to swing too far in the opposite direction either. The goal shouldn't be to reject AI altogether out of fear, but to build a culture where these systems are properly tested, explained and held accountable before they're trusted with high-stakes decisions. Trust that is earned through transparency and evidence is much more durable than trust that is simply assumed because a system "seems" advanced or scientific.



Task 3: Measures to Address the Issue

After looking into the COMPAS case and the ethical issues from Task 1, I've grouped the possible solutions into a few categories — technical measures that developers can apply directly, and organisational or policy measures that need support from companies, institutions and governments.

Technical Measures

- Use more diverse and representative training data, and actually audit datasets for imbalance before training rather than assuming the data is "fine" by default.
- Run regular bias and fairness testing on models, both before and after deployment, using established fairness metrics (such as equalised odds or demographic parity) rather than checking only once at launch.
- Use explainability tools, like SHAP or LIME, so that the reasoning behind a model's decision can actually be inspected by humans, rather than being trusted blindly as a black box.
- Apply privacy-preserving methods like differential privacy or federated learning, which allow models to learn useful patterns without directly exposing or storing individuals' raw personal data.

Organisational and Policy Measures

- Keep a human in the loop for high-stakes decisions — for example, a judge, doctor or hiring manager should treat an AI's output as one input among several, not the final word.
- Push for outside, independent audits of systems used in areas like courts, hiring or lending, so they are open to scrutiny instead of being a black box controlled entirely by one company.
- Support clearer regulation and disclosure requirements — similar to what's being proposed under the EU AI Act — so that high-risk AI systems used in the public sector cannot stay completely hidden from public accountability.
- Keep monitoring models after they're deployed, since bias and accuracy can drift over time as real-world data changes; ethical AI isn't something you fix once and then forget about.

Challenges in Implementing These Measures

It's worth being honest that none of these measures are simple to put into practice. Collecting more representative data, for instance, can be expensive and time-consuming, and in some cases the historical data available simply doesn't exist in a fairer form — the bias is baked into the only records we have. Similarly, explainability tools like SHAP can tell us which features influenced a decision, but they don't always give a full, intuitive explanation that a non-technical person, like a defendant in a courtroom, could easily understand.

There's also a genuine tension between accuracy and fairness in some cases: forcing a model to satisfy a strict fairness constraint can sometimes reduce its overall predictive accuracy, which creates a difficult trade-off for organisations to navigate, especially when regulation hasn't clearly decided which value should take priority.

None of this means the measures aren't worth pursuing; it just means that fixing these issues is an ongoing process rather than a one-time technical patch, and it needs cooperation between engineers, companies, regulators and the public rather than being left to any single group alone.

Conclusion

Working on this project made me realise that ethical issues in deep learning aren't just abstract, philosophical debates — they show up in real systems that affect real people, like in the COMPAS case. Bias, privacy and transparency all tie back to one core idea: powerful systems need to be built and monitored carefully, with real accountability, so that the people affected by them are actually treated fairly, and so that trust in AI is genuinely earned rather than just assumed.

As deep learning keeps getting used in more parts of everyday life, I think the responsibility to get this right will only keep growing, and it's going to take input from far more than just computer scientists — lawyers, ethicists, affected communities and regulators all need a seat at the table.